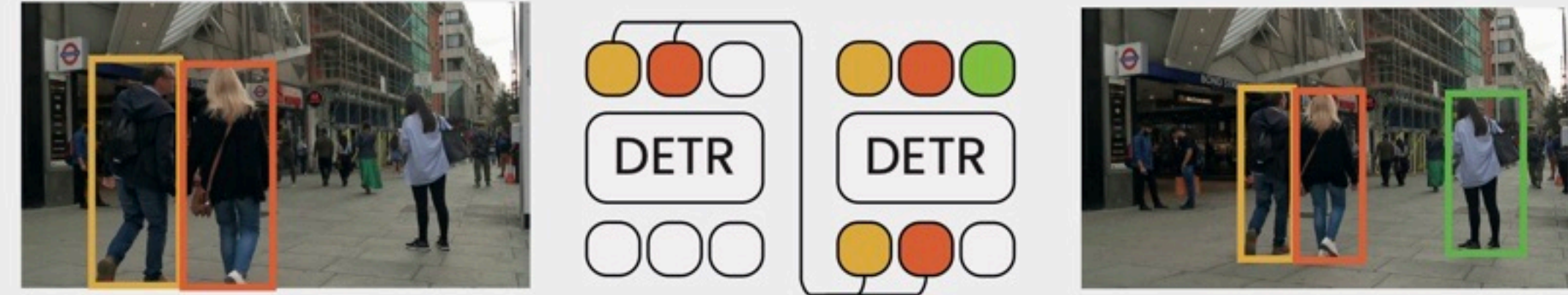


1. Intro - Multiple Object Tracking

Multiple Object Tracking (MOT) seeks to continuously detect and identify objects in a sequence of frames, i.e., a video.

Tracking-by-detection: detection and tracking are handled independently by solving the hungarian algorithm.

Tracking-by-query: jointly perform detection and tracking inherently in an iterative **end-to-end** attention mechanism.



2. Problem statement: domain-shifts

End-to-end trackers learn to track completely from annotated video data. However, these trackers are **data-hungry**.

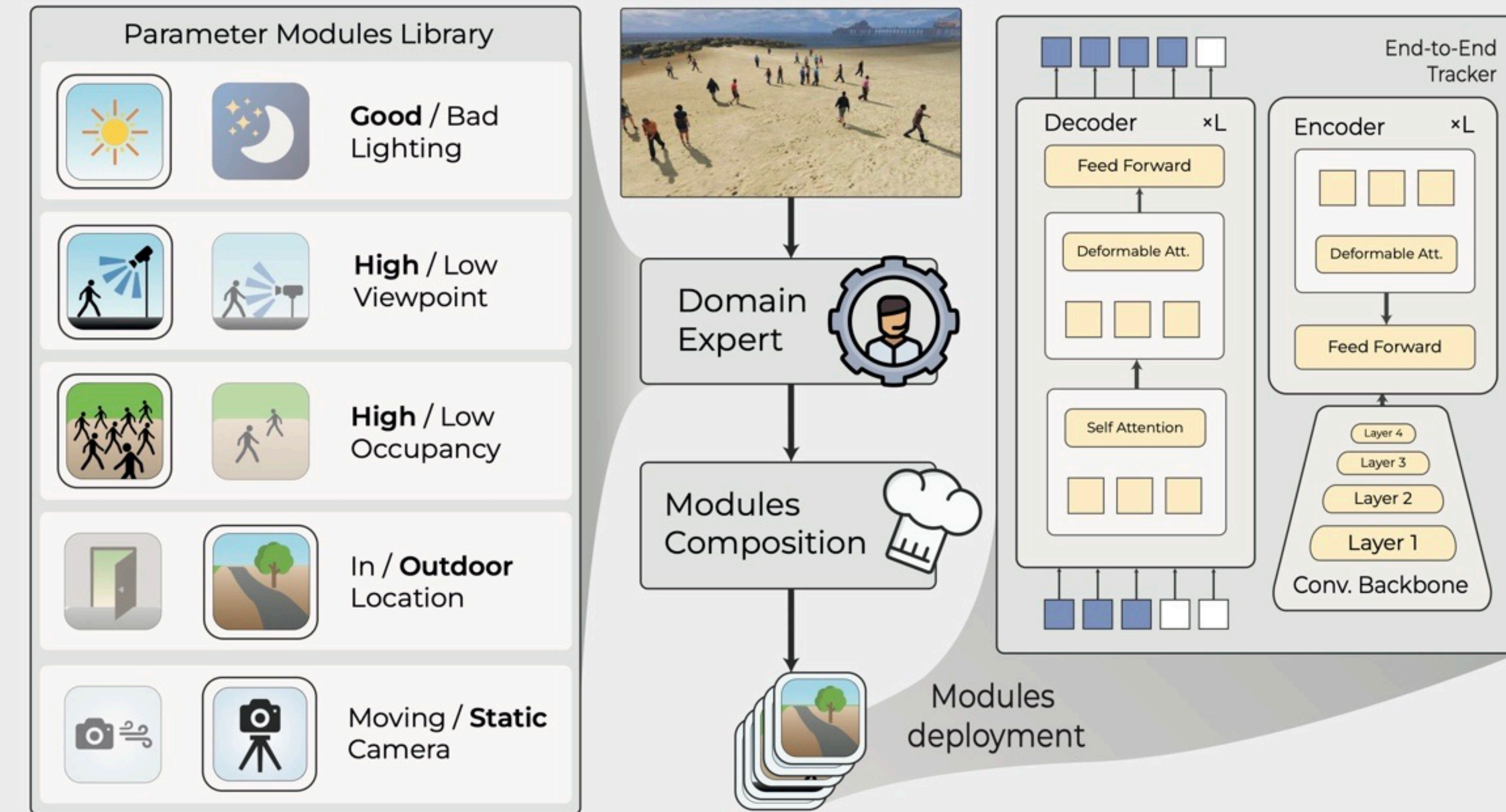
The limited availability of annotated data often leads end-to-end trackers to overfit on training sets, making them **vulnerable to domain shifts** [1].

With limited data, the model struggles to generalize during training, especially when **negative interference** between scenarios with differing **attributes** arises.



3. Solution: attribute-specific PEFT modules

Idea: During training we train a **parameter-efficient module** (e.g, LoRA [2]) for each attribute. During inference we compose expert modules for each scenario selected by a domain expert.



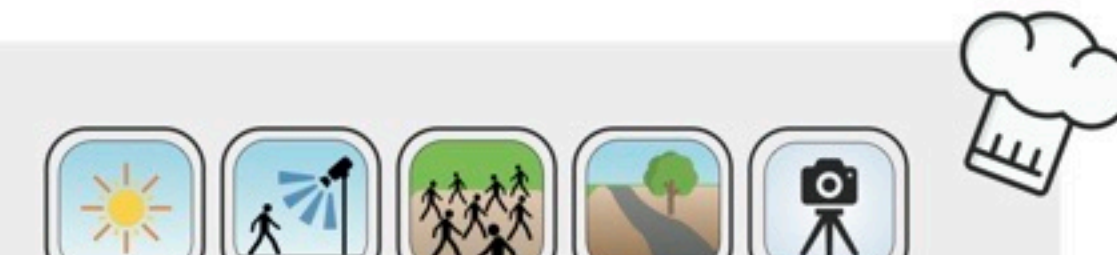
4. Modules composition

Parameter-efficient Scenario-specific Tracking Architecture

We train each **PEFT** [2] module independently and we compose these modules during inference:

- (Eq. 1) Modules are defined as task vectors
- (Eq. 2) Encoder-decoder modules are **LoRA** units.
- (Eq. 3) We define $\rho < 1$ in domain-shifts to weigh unselected modules.

→ **Domain-shift:** we leverage even the unselected modules, ensuring **no knowledge is discarded** [3].



(1)

$$\theta_c = \theta_0 + \sum_i \lambda_i \tau_i$$

(2)

$$\tau_i = \sum_{m \in R(i)} \bar{\lambda}_m B_m A_m$$

(3)

$$\bar{\lambda}_m = \begin{cases} \rho & \text{if } m \text{ is selected} \\ \frac{1-\rho}{|R(i)|-1} & \text{otherwise} \end{cases}$$

5. Experimental results

Zero-shot experiments

Starting from the MOTSynth training, we test PASTA on real datasets (**MOT17**, **PersonPath22**) to validate the improved generalization achieved by composing expert modules, in a zero-shot evaluation.

Method	MOT17			PersonPath22		
	HOTA↑	IDF1↑	MOTA↑	MOTA↑	IDF1↑	IDSW↑
MOTRv2-MS	62.6	73.0	67.6	48.3	53.1	7154
PASTA ($\rho = 1$)	63.7	74.1	67.9	49.7	53.7	6321
PASTA ($\rho = 0.8$)	64.0	74.9	68.1	50.0	53.8	6037

In-domain experiments

We train and test PASTA on the synthetic dataset (**MOTSynth**) to show how reducing negative interference enhances association metrics.

Method	$ \Theta $	HOTA↑	IDF1↑	MOTA↑	DetA↑	AssA↑
ByteTrack	-	45.7	56.4	61.8	50.1	41.9
MOTRv2-MS	42M	52.4	56.6	61.9	56.4	49.0
PASTA (Ours)	15M	53.0	57.6	62.0	56.2	50.4

Ablation studies

We validate that in zero-shot is better not to discard any knowledge.

Method	In-domain			Domain-shift		
	HOTA↑	IDF1↑	MOTA↑	HOTA↑	IDF1↑	MOTA↑
Weighted avg. (all)	59.9	66.8	59.6	64.0	74.9	68.1
Avg. (only selected)	60.1	67.2	59.9	63.7	74.1	67.9

[1] Zhang et al. "Motrv2: Bootstrapping end-to-end multi-object tracking by pretrained object detectors." CVPR 2023.

[2] Hu et al, "Lora: Low-rank adaptation of large language models", ICLR 2022

[3] Zhang et al. "Learning useful representations for shifting tasks and distributions." ICML, 2023.

